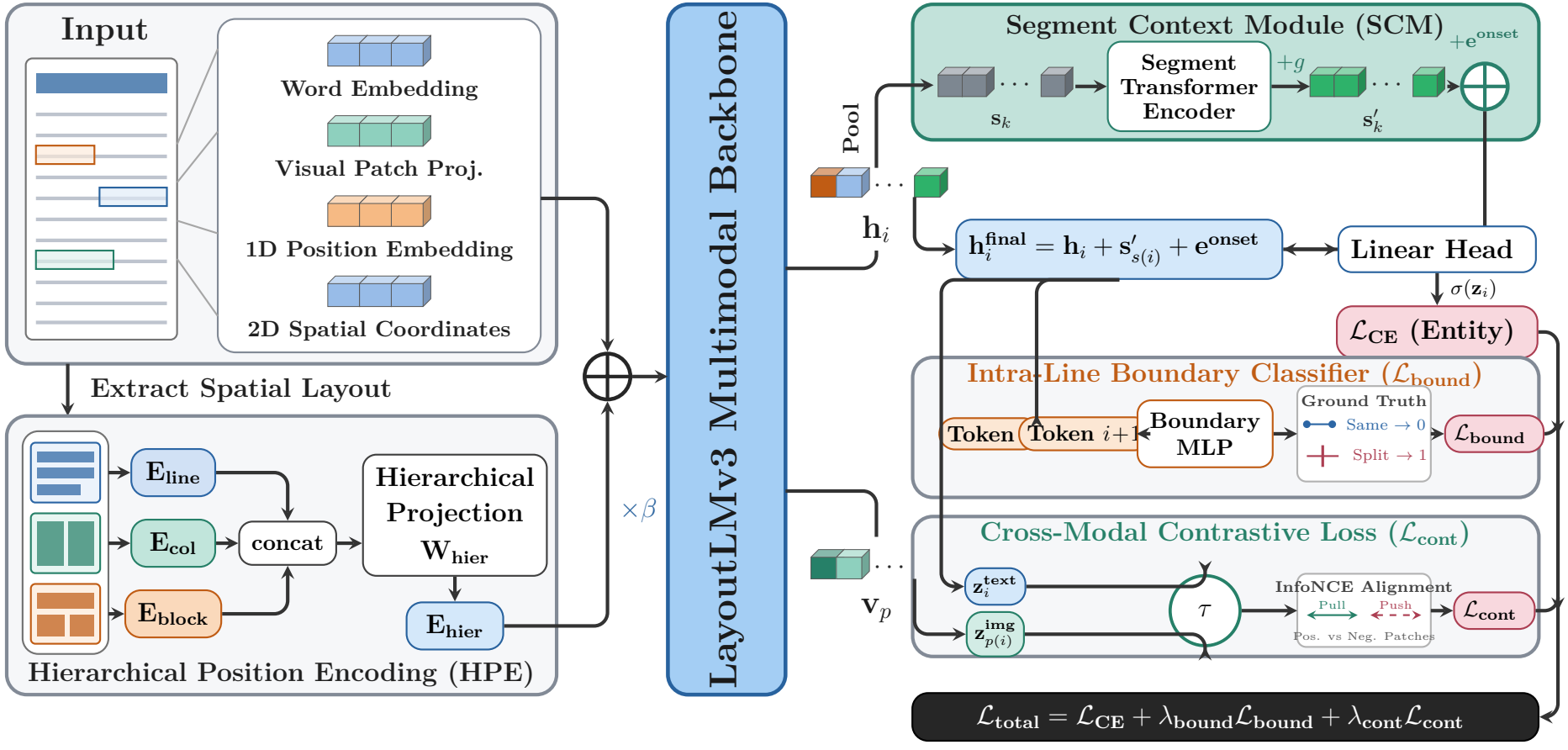
Input
Word Embedding
Visual Patch Proj.
1D Position Embedding
2D Spatial Coordinates
Extract Spatial Layout
$\mathbf{E}_{\text{line}}$
$\mathbf{E}_{\text{col}}$
$\mathbf{E}_{\text{block}}$
concat
Hierarchical Projection $\mathbf{W}_{\text{hier}}$
$\mathbf{E}_{\text{hier}}$
Hierarchical Position Encoding (HPE)
$\times\beta$
LayoutLMv3 Multimodal Backbone
Pool
$\mathbf{h}_i$
$\mathbf{v}_p$
Segment Context Module (SCM)
$\mathbf{s}_k$
Segment Transformer Encoder
$+g$
$\mathbf{s}'_k$
$+\mathbf{e}^{\text{onset}}$
$\mathbf{h}_i^{\text{final}} = \mathbf{h}_i + \mathbf{s}'_{s(i)} + \mathbf{e}^{\text{onset}}$
Linear Head
$\sigma(\mathbf{z}_i)$
$\mathcal{L}_{\text{CE}}$ (Entity)
Intra-Line Boundary Classifier ($\mathcal{L}_{\text{bound}}$)
Token
Token $i+1$
Boundary MLP
Ground Truth
Same $\rightarrow$ 0
Split $\rightarrow$ 1
$\mathcal{L}_{\text{bound}}$
Cross-Modal Contrastive Loss ($\mathcal{L}_{\text{cont}}$)
$\mathbf{z}_i^{\text{text}}$
$\mathbf{z}_{p(i)}^{\text{img}}$
τ
InfoNCE Alignment
Pull
Push
Pos. vs Neg. Patches
$\mathcal{L}_{\text{cont}}$
$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{bound}}\mathcal{L}_{\text{bound}} + \lambda_{\text{cont}}\mathcal{L}_{\text{cont}}$